

Prepared By	09-Jun-2012	Mr.Chatchawan S.	 Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnauy W.		
Certified By	09-Jun-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved
1	13-Feb-2013	Mr.Chatchawan S.	Mr.Phuchin S.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-002-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-005-1**



WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**
WPS No. **RNC-WPS-005-1** Date **13-Feb-13** Revision No. **1**
Supporting PQR No.(s) **RNC-PQR-005-1** Date **13-Feb-13** Type(s) **Manual**
Welding Process(es) **Gas Tungsten Arc Welding (GTAW)** (Automatic, Manual, Machine, or Semi-Auto.)

JOINT (QW-402)

Joint Design **Single V**
Backing (Yes) ☐ (No) ☒ X
Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **1** Group No. **1** to P-No. **1** Group No. **1**
Material Specification Type and Grade **SA 333 Gr.1.6 SA420 WPL 6** TO **SA 333 Gr.1.6 SA420 WPL 6**
Chem. Analysis and Mech. Prop.
Thickness Range :
Base Metal : Groove **0.5 mm - 12.04 mm.** Fillet **All**
Pipe Dia. Range : Groove **All** Fillet **All**
Other

FILLER METALS (QW-404)

	GTAW
Spec. No. (SFA)	5.28
AWS No. (Class)	ER-70S-G
F-No.	6
A-No.	1
Size of Filler Metals	Ø 2.0 & 2.4 mm
Deposited Weld Metals	6.02 mm
Thickness Range :	
Groove	12.04 mm Max.
Fillet	All
Electrode-Flux (Class)	-
Flux Trade Name	-
Consumable Insert	-
Other	KobeTGS-1N or Equivalents

POSITION (QW-405)

Position(s) of Groove **ALL**
Welding Progression **Up** ☒ **Down** ☐
Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **N/A**
Time Range **N/A**
Notes

PREHEAT (QW-406)

Preheat Temp. Min. **10°C**
Interpass Temp. Max. **238°C**
Preheat Maintenance **N/A**
Notes

GAS (QW-408)

	Percent Composition		
	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	15-20Ltr/Min
Trailing	-	-	-
Backing	-	-	-

WPS No. RNC-WPS-005-1 Rev. 1

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☐ AC ☒ DC Polarity EN
 Amps (Range) 80-150 Volts (Range) 9-13
 Tungsten Electrode Size and Type Ø 2.4 mm, Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

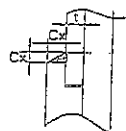
String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size 8.0 mm - 19.0 mm
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) 5.0-8.0Cm/Minute
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER70S-G	2.4	DC EN	80-100	10-12	5.0-7.0	
2nd & Other(s)	GTAW	ER70S-G	2.4	DC EN	90-142	11-13	6.0-8.0	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of ASME-IX, 2007 latest Revision.

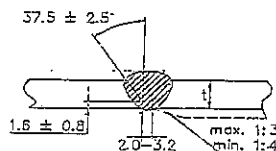
Prepared by: Mv. Chatchawan S. 13/02/13
 Name Signed Date
 Reviewed by: ภูชิน สีหอนไชย 22/03/13
 Name Signed Date
 Approved by: ภูชิน สีหอนไชย 22/03/13
 Name Signed Date

ATTACHED FIGURE JOINT SKETCH



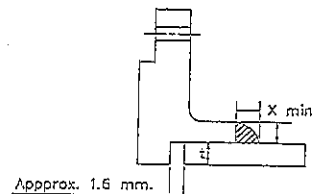
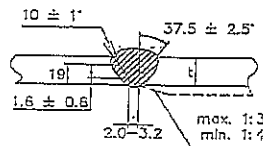
C_x (min.) = $1.25t$ but not less than 3.2 mm.

JOINT No. A1



Misalignment (max.) = 1.5 mm.

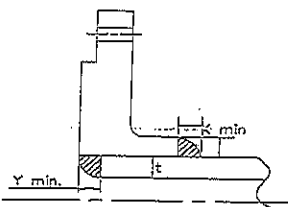
JOINT No. B1 & B2



Approx. 1.6 mm.

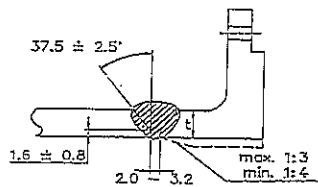
X min. = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



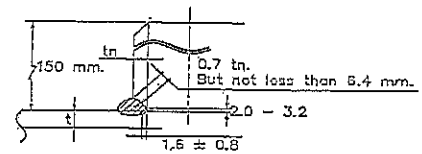
X min. = The lesser of $1.4t$ or thickness of the hub.
Y min. = The lesser of t or 5.4 mm.

JOINT No. A3



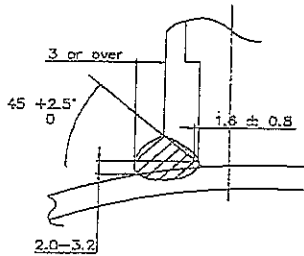
Misalignment (max.) = 1.5 mm.

JOINT No. B3

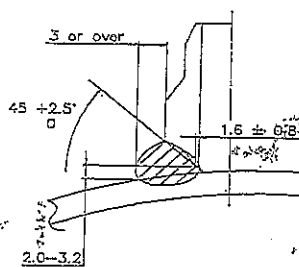


Misalignment (max.) = The lesser of 3.2 mm. or $0.5t_n$.

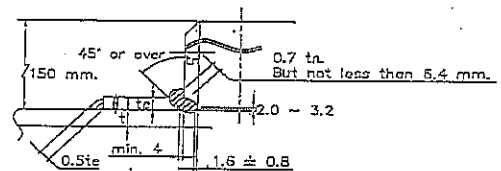
JOINT No. B4



JOINT No. C1

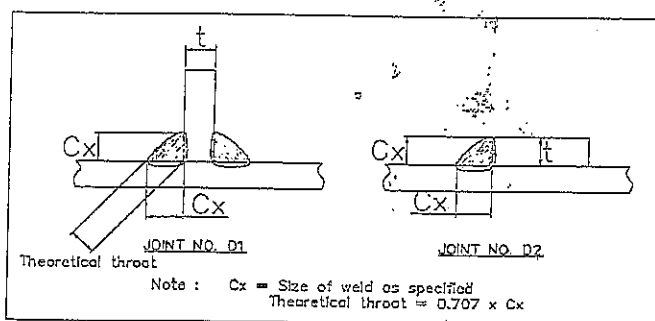


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or $0.5t_n$.

JOINT No. B5





Procedure Qualification Record (PQR)

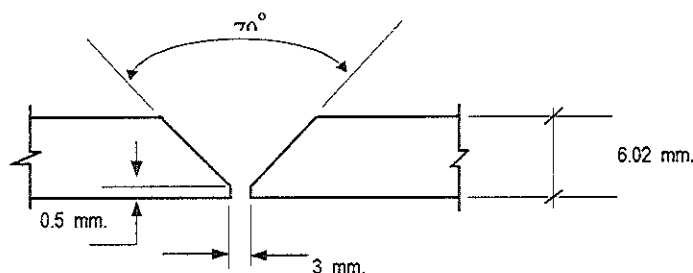
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-005-1

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-005-1	Date :	13 February 2013
WPS Number :	RNC-WPS-005-1		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)																			
Material Spec.	A 333	To	A 333	Temperature	N/A																		
Type or Grade	Gr.6	To	Gr.6	Hold Time	N/A																		
P. No.	1	To	1	Others	GAS (QW-408) Percent Composition <table><tr><td></td><td>Gas (es)</td><td>Mixture</td><td>Flow Rate</td></tr><tr><td>Shielding</td><td>Argon</td><td>99.99%</td><td>15-20 L/Min.</td></tr><tr><td>Trailing</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Backing</td><td>-</td><td>-</td><td>-</td></tr></table>				Gas (es)	Mixture	Flow Rate	Shielding	Argon	99.99%	15-20 L/Min.	Trailing	-	-	-	Backing	-	-	-
	Gas (es)	Mixture	Flow Rate																				
Shielding	Argon	99.99%	15-20 L/Min.																				
Trailing	-	-	-																				
Backing	-	-	-																				
Thickness of Test Coupon	6.02 mm.																						
Diameter of Test Coupon	Ø 4"																						
Others																							
FILLER METALS (QW-404)				ELECTRICAL CHARACTERISTICS (QW-409)																			
SFA Specification	A 5.28			Current	DC																		
AWS Classification	ER70S-G			Polarity	EN																		
Filler Metal F-No.	6			Amperage	85 - 142	Voltage	9 - 12																
Weld Metal Analysis A-No.	1			Tungsten Electrode Size	Ø 2.4 mm. 2% thoriated.EWTh-2																		
Size of Filler Metal	Ø 2.4 mm.			Others	N/A																		
Others	Kobelco TGS-1N			TECHNIQUE (QW-410)																			
Weld Metal Thickness	-			Travel Speed	5.04 - 7.43 cm/min.																		
POSITION (QW-405)				String or Weave Bead	Both																		
Position of Groove	6G			Oscillation	N/A																		
Progression of Welding (Uphill or Down)	Up Hill			Multiple or single Pass	Multiple Pass																		
Others	N/A			Multiple or Single Electrodes	Single Electrode																		
PREHEAT (QW-406)				Others	N/A																		
Preheat Temperature	33° C																						
Interpass Temperature (max.)	189° C																						
Others	N/A																						

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- RNC-PQR-005-1

Tensile Test Report No.TS-13-02-088

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
549/13 TR1	19.20	5.69	109.25	51.83	474.41	OUT OF WELD
549/13 TR2	19.17	5.83	111.76	53.07	474.87	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-02-037

Type and Figure Number	Type of Bend	Indication	Results
549/13 BF1	Transverse face bend	No Open Discontinuity	Accepted
549/13 BF2	Transverse face bend	No Open Discontinuity	Accepted
549/13 BR1	Transverse root bend	No Open Discontinuity	Accepted
549/13 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.WQT-004

Hardness Testing Report No. HV-13-02-022

Macrostructure Report No. MA-13-02-056

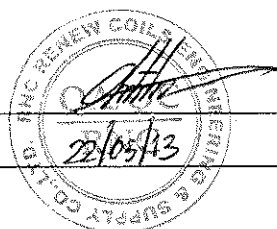
Impact Testing Report No.IM-13-03-031 (-60°C)

Welders Name Mr.Pongpet H. Welder Number W-002
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date :



Inspected by :

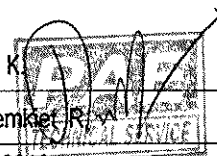
Certified by :

Date :

Mr.Wichit K.

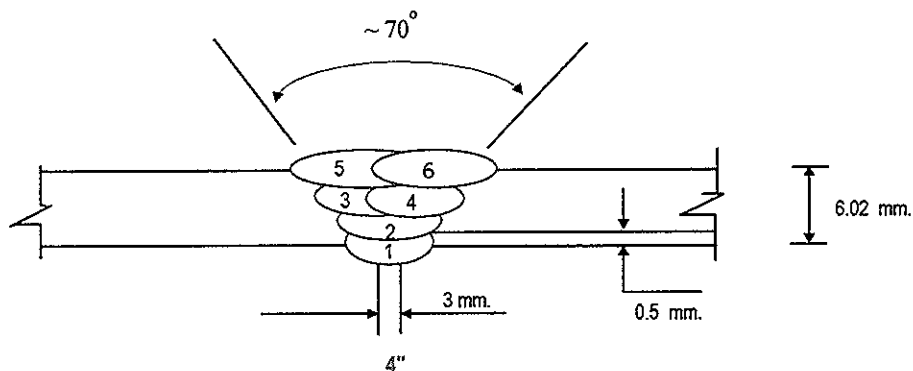
Mr.Chaloemkit R.

19 March 2013



PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : RNC-PQR-005-1
		WPS No. : RNC-WPS-005-1
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST CONDITION AND RESULT

Welding Sequences for PQR-RNC-005-1 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or	Wire Feed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Weave Bead	Speed				
1	GTAW	ER70S-G	2.4	85	9	DCEN	-	-	5.11	8.99	20	-
2	GTAW	ER70S-G	2.4	115	10	DCEN	-	-	7.43	9.29	20	-
3	GTAW	ER70S-G	2.4	142	12	DCEN	-	-	6.28	16.28	20	-
4	GTAW	ER70S-G	2.4	140	12	DCEN	-	-	5.78	17.44	20	-
5	GTAW	ER70S-G	2.4	126	11	DCEN	-	-	5.04	16.50	15	-
6	GTAW	ER70S-G	2.4	130	11	DCEN	-	-	7.21	11.90	15	-





PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

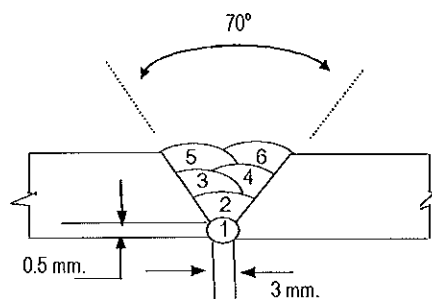
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : PQR		Location : RNC Work Shop		Report No. PQR-015	
		Customer Order No. : N/A		WPS No. : RNC-WPS-005-1		Date : 13-Feb-13	

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
W-002	MR.PONGPET H.	GTAW	6G	ASME IX	A333Gr.6	ø 4" Thk. 6.02 mm.	A5.28	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	DCEN	85	9	5.11	33	8.99	Argon	20	-	-
2	GTAW	ER70S-G	2.4	DCEN	115	10	7.43	152	9.29	Argon	20	-	-
3	GTAW	ER70S-G	2.4	DCEP	142	12	6.28	181	16.28	Argon	20	-	-
4	GTAW	ER70S-G	2.4	DCEP	140	12	5.78	185	17.44	Argon	20	-	-
5	GTAW	ER70S-G	2.4	DCEP	126	11	5.04	190	16.50	Argon	15	-	-
6	GTAW	ER70S-G	2.4	DCEP	130	11	7.21	189	11.90	Argon	15	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :

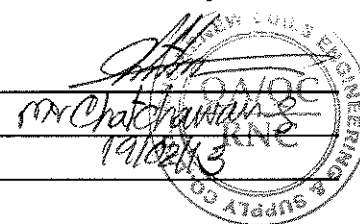
Mr.Wichit K

14-Feb-2013

SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)607402-3, Fax : 038)607402, SRIRACHA : 038)330220, 330229

Page : 1 Of 1

REPORT No.	WQT-004	RADIOGRAPHIC EXAMINATION REPORT						INSPECTION & TEST RECORDS	
DWG.No.	-								
DATE :	13-Feb-13								
CLIENT : Renew Coils Engineering & Supply Co.,Ltd.							PROJECT NAME : Welder Supply		
PROCEDURE No.:	PAE-RT-001 (Rev.0)	EXPOSURE TIME	4 min.	10 sec.	DISTANCE		4" Inch. mm.		
RADIATION SOURCE	X-ray (-) MAX - KVG. γ ray (Ir-192)	DENSITY RANGE	2.0 - 3.5		TYPE OF PENETRATOR		ASTM-1A		
SOURCE RANGE	7.32 Ci.	FILM TYPE SIZE	KODAK "MX" (Type I)	FILM PROCESSING		AUTO MANUAL	-		
SOURCE SIZE	3 x 3 mm.		3.5"x 8.5"			✓			
TUBE VOLTAGE	- KVP.	INTENSIFYING SCREEN	FRONT BACK	0.005 0.005	Inch. Inch.	MATERIAL THICKNESS	A333 Gr.6 6.02 mm.		
TUBE CURRENT	- mA								
APPLICABLE STANDARD : ASME Section V Article 2						ACCEPTANCE CRITERIA : ASME-IX			
RADIOGRAPHIC TECHNIQUE :		☐ SWSI ☒ DWSI ☐ DWDI							
NAME / WQT	Position No.	Pipe/Plate Size		Welder No.	Film No.	Interpretation	Results	Remarks	
		Dia	Thk(mm).						
PQR / GTAW / WPS No.RNC-005-1									
MR.PONGPET H.	6G	4"	6.02	W-002	A-B	NAD	ACC		
					B-C	NAD	ACC		
					C-A	NAD	ACC		
Total						3.5"x8.5"	3	Films	
						3.5"x17"	-	Films	
ABBREVIATIONS :			C : Crack			LF : Lack Of Fusion			
ACC : Accepted			P : Porosity			CRP : Concave Root Penetration			
REJ : Rejected			UC : Undercutting			ERP : Excess Root Penetration			
SF : Surface Flaw			SL : Slag Inclusion			NAD : No Apparent Defects			
			BT : Burn Through			RS : Reshoot			
PAE INSPECTED BY				REVIEWED BY				APPROVED BY	
SIGNATURE :				SIGNATURE :				SIGNATURE :	
NAME : Mr.Sarawat P.				NAME :				NAME :	
DATE : 14-Feb-13				DATE :				DATE :	



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnplm.co.th

29784

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-02-088

Revision No. 0

Test Product	WPS No. RNC-WPS-005-1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A333 Gr.6	Type of Test	Reduced Section Tension
Dimension	Ø 4" (6.02 mm Thk.)	Equipment's Capacity	100 TONS
Date Received	19 February 2013	Equipment/ Serial No.	INSTRON / 1000DXR1414
Date Tested	20 February 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	549/13 TR1	N/A	5.69	19.20	109.25	N/A	N/A	N/A	51.83	474.41	N/A	OUT OF WELD
2	549/13 TR2	N/A	5.83	19.17	111.76	N/A	N/A	N/A	53.07	474.87	N/A	OUT OF WELD

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Pongpet H.

ID No. 3 2512 00585 27 8

Welder No. : W-002

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.

3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.

4. This report is tested for LPN Metallurgical Research Center (Thailand) Co.,Ltd.



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th

29790

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-02-037

Revision No. 0

Test Product	WPS No. RNC-WPS-005-1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A333 Gr.6	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 4" (6.02 mm Thk.)	Diameter of Meudrel	22 mm
Date Received	19 February 2013	Bend Angle	180 Degree.
Date Tested	20 February 2013	Equipment/ Serial No.	SHIMADZU UH200A/32098926

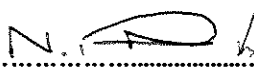
Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	549/13 BF1	Transverse face bend	5.70x19.06x161.58 mm	No Open Discontinuity	N/A	-
2	549/13 BF2	Transverse face bend	5.70x19.01x162.03 mm	No Open Discontinuity	N/A	-
3	549/13 BR1	Transverse root bend	5.75x19.07x161.86 mm	No Open Discontinuity	N/A	-
4	549/13 BR2	Transverse root bend	5.75x19.06x160.45 mm	No Open Discontinuity	N/A	-

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Pongpet H.

ID No. 3 2512 00585 27 8

Welder No. : W-002

Reported by : 

(Mr. Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by : 

(Mr. Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

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LPN PLATE MILL



Testing Laboratory LPN Plate Mill Public Company Limited

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13236

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Report No. HV-13-02-022

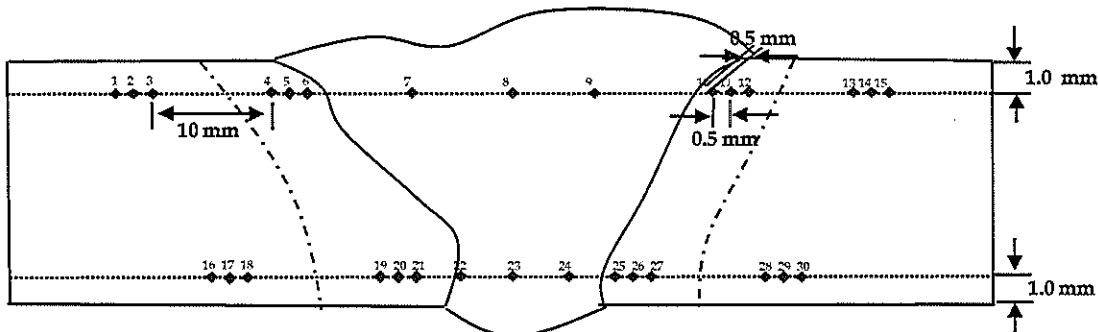
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000 Thailand.

Revision No. 0

Test Product	WPS No. RNC-WPS-005-1	Material Characterization	Pipe
Test No.	549/13 HN	Test Method	ASTM E92-82 (Reapproved 2003) ^{E2}
Project Name	WELDER SUPPLY AND PQR	Standard of	ASME Section IX : 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A333 Gr.6	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	4" (6.02 mm Thk.)	Indenter Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW/6G	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	18 February 2013	Cal. Block	No. A60023 No. A62072
Date Tested	20 February 2013	Ambient Temperature	26.0 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)
1	BASE	133.7	16	BASE	134.2			
2	BASE	133.6	17	BASE	133.8			
3	BASE	135.5	18	BASE	134.8			
4	HAZ	144.7	19	HAZ	145.4			
5	HAZ	147.4	20	HAZ	149.5			
6	HAZ	156.5	21	HAZ	151.2			
7	WELD	152.5	22	WELD	148.5			
8	WELD	167.9	23	WELD	149.2			
9	WELD	171.5	24	WELD	146.5			
10	HAZ	160.7	25	HAZ	153.4			
11	HAZ	154.7	26	HAZ	153.0			
12	HAZ	142.3	27	HAZ	149.6			
13	BASE	129.7	28	BASE	131.2			
14	BASE	128.0	29	BASE	133.4			
15	BASE	131.6	30	BASE	132.8			

Additional details :-

Reported by :
(Mr. Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 22/2/13



Approved by :
(Mr. Pavaret Freedawiphat)
Deputy Managing Director
Date : 22/2/13

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Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

13700

Macrostructure Analysis Report

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

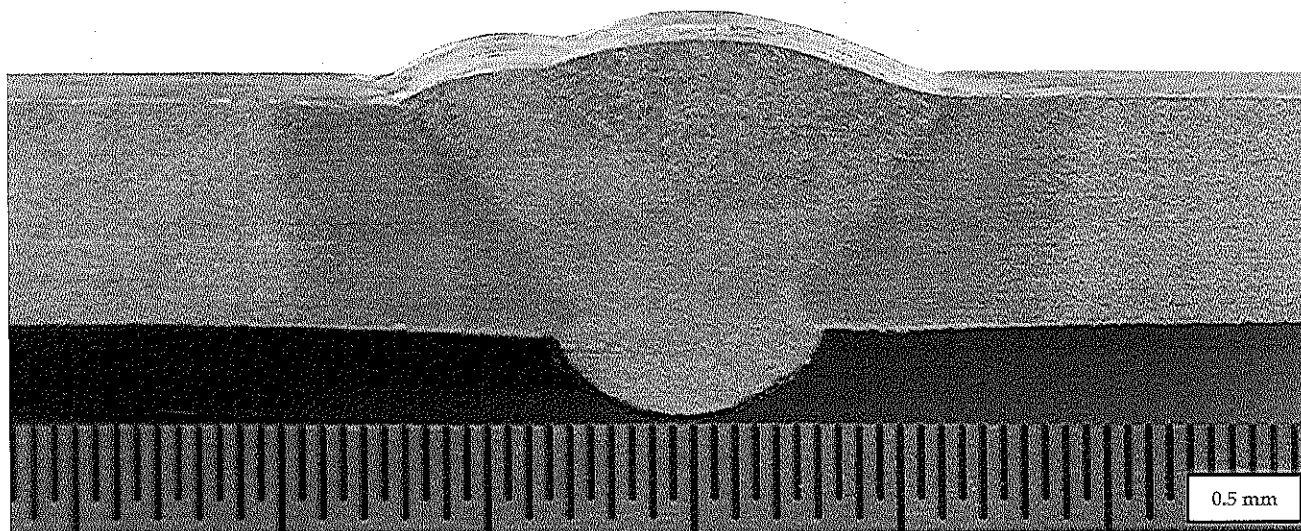
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Page 1 of 1

Report No. MA-13-02-056

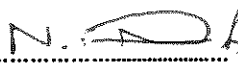
Revision No.0

Test Product	WPS No. RNC-WPS-005-1	Material Characterization	Pipe
Test No.	549/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	A333 Gr.6	Method of Preparation	ASME Section IX : 2010 edition
Dimension	4" (6.02 mm Thk.)	Test Method	ASME Section IX : 2010 edition
Welding Process/Position	GTAW/6G	Standard of Test	ASME Section IX : 2010 edition
Type of Joint	Butt Jiont	Etchant Reagent	25 ml HNO ₃ + 75 ml H ₂ O
Welder Name / No.	Mr. Pongpet H. / W-002	Time of Etching	5 minutes
Date Received	18 February 2013	Date Tested	20 February 2013



Macro-etch Examination : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

Additional details : -

Reported by : 
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 22/02/13



Approved by : 
(Mr.Pavaret -Preedawiphat)
Deputy Managing Director
Date : 22/02/13

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28668

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnplm.co.th

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-03-031

Revision No. 0

Test Product	WPS No. RNC-WPS-005-1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 and ASTM E23M-07a ^{cl}
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A333 Gr.6	Type of Test	Charpy V-Notch Impact
Dimension	Ø 4" (6.02 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Received Date	18 February 2013	Equipment 's Capacity	542 Joules
Tested Date	18 March 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	549/13-1	-60°C	3.310	10.016	54.89	22.21	19.68	Base Metal	-
2	549/13-2	-60°C	3.301	10.016	54.84	17.98			
3	549/13-3	-60°C	3.302	10.021	54.77	18.86			
4	549/13-1	-60°C	3.309	10.020	54.84	54.21	55.89	HAZ	-
5	549/13-2	-60°C	3.310	10.018	54.80	50.98			
6	549/13-3	-60°C	3.303	10.019	54.86	62.47			
7	549/13-1	-60°C	3.309	10.020	54.73	62.47	58.08	Weld Metal	-
8	549/13-2	-60°C	3.307	10.018	54.74	57.45			
9	549/13-3	-60°C	3.302	10.016	54.70	54.31			

Additional details : Welding Process / Position : GTAW / 6G

Welder Name : Mr. Pongpet H.

ID No. 3 2512 00585 27 8

Welder No. : W-002

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 19 / 03 / 13

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

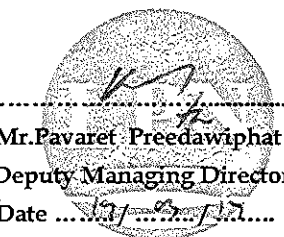
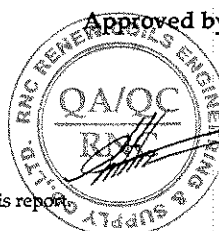
Date 19 / 03 / 13

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